Mathematical and Computational Methods

Methods

Part II: Vectors and matrices (Units 4–6)

Most problems on the exercise sheets are instances of a few standard types, and each type has a *method*: a fixed sequence of steps, taken from the notes, that solves every problem of that type. This guide collects them, unit by unit. Each method states the problem it solves, lists its steps and works one example that is not on any sheet. It ends with the mistakes to avoid, where the notes teach it, and a link to a Review set of problems of that kind. The units fall into four parts: foundations and complex numbers, vectors and matrices, differentiation, and integration.

The guide does not say which sheet problem needs which method. Recognising the type is part of what the sheets train, and in an exam nothing tells you. Where a unit's problems look alike but need different methods, its chapter opens with a table for choosing.

Contents

Part II Vectors and matrices 2

Unit 4 Vectors . . . 2
- 4.1 Working in components: length, unit vectors, combinations . . . 2
- 4.2 The angle between vectors, and between planes . . . 3
- 4.3 Projection, resolving, work . . . 4
- 4.4 The cross product: normal, area, volume, torque . . . 5
- 4.5 A line through two points; is a point on it? . . . 6
- 4.6 A plane through three points . . . 7
- 4.7 Where a line meets a plane; distance from a point to a plane . . . 8

Unit 5 Matrices . . . 11
- 5.1 Products, transpose and trace: check the sizes first . . . 11
- 5.2 Determinants by cofactor expansion; the parameter that makes a matrix singular . . . 12
- 5.3 The inverse: the 2×2 formula, cofactors, or row reduction . . . 13
- 5.4 Gaussian elimination and reading the echelon form . . . 15
- 5.5 A matrix as a transformation of the plane . . . 16

Unit 6 Eigensystems . . . 18
- 6.1 The eigensystem: 2×2, 3×3, complex eigenvalues . . . 18
- 6.2 Is it diagonalisable? Multiplicities . . . 19
- 6.3 Diagonalising, and powers of a matrix . . . 20
- 6.4 A recurrence, by decomposing the start vector . . . 21
- 6.5 Symmetric and Hermitian matrices: orthogonal diagonalisation, PCA . . . 23
- 6.6 A Markov chain's long run . . . 24

Part II Vectors and matrices

Units 4–6: vectors, matrices, and the eigenvalues and eigenvectors of a matrix. The methods build on one another. The cross product gives a plane its normal, row reduction gives both an inverse and an eigenvector, and the powers of a matrix come from its eigenvectors.

Unit 4 Vectors

Seven methods: the algebra of components first, then three on the two products and three on lines and planes. Proofs from the dot-product rules are a variant of Method 4.2.

4.1 Working in components: length, unit vectors, combinations

Problem. Add and scale vectors; find the vector between two points and the distance between them; normalise a vector; write a vector as a linear combination of two others, and say whether those two are independent.

Method

1. For the vector between two points, take end minus start: $\overrightarrow{PQ} = (q_1 - p_1,\ q_2 - p_2,\ q_3 - p_3)$.
2. Add, subtract and scale componentwise. Scale each vector first, then combine them component by component.
3. The length is $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$. The distance from P to Q is $\left\|\overrightarrow{PQ}\right\|$.
4. To normalise $\mathbf{v} \neq \mathbf{0}$, divide by its length: $\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$. Check that $\|\hat{\mathbf{v}}\| = 1$.
5. To write $\mathbf{c} = \alpha\mathbf{a} + \beta\mathbf{b}$, match the components. That gives two equations in α and β. Solve them by substitution, and check by recombining.
6. Two vectors are independent exactly when neither is a multiple of the other. Try $\mathbf{b} = \lambda\mathbf{a}$: if the components need different values of λ, the two vectors are independent. Two independent vectors in $\mathbb{R}^2$ form a basis: every vector is a combination of them in exactly one way.

Example. (a) For $P(1, 4, -2)$ and $Q(3, -2, 1)$, find $\overrightarrow{PQ}$, the distance PQ, and the unit vector in the direction from P to Q. (b) Write $\mathbf{c} = (8, -3)$ as a combination of $\mathbf{a} = (2, 1)$ and $\mathbf{b} = (1, -3)$. Are $\mathbf{a}$ and $\mathbf{b}$ independent?

(a) End minus start, $\overrightarrow{PQ} = (3 - 1,\ -2 - 4,\ 1 - (-2))$, so

$$\boxed{\overrightarrow{PQ} = (2, -6, 3)}$$

Then $\left\|\overrightarrow{PQ}\right\| = \sqrt{4 + 36 + 9} = \sqrt{49}$, so $\boxed{PQ = 7}$. With $\mathbf{v} = \overrightarrow{PQ}$, dividing by 7 gives

$$\boxed{\hat{\mathbf{v}} = \left(\tfrac{2}{7}, -\tfrac{6}{7}, \tfrac{3}{7}\right)}$$

(b) Matching components gives $2\alpha + \beta = 8$ and $\alpha - 3\beta = -3$. The first equation gives $\beta = 8 - 2\alpha$. Substituting, $\alpha - 3(8 - 2\alpha) = -3$, so $7\alpha = 21$. Hence $\alpha = 3$ and $\beta = 2$:

$$\boxed{\mathbf{c} = 3\mathbf{a} + 2\mathbf{b}}$$

For independence, suppose $\mathbf{b} = \lambda\mathbf{a}$. The first components need $\lambda = \frac{1}{2}$, and the second need $\lambda = -3$. (Likewise $\mathbf{a} = \mu\mathbf{b}$ needs $\mu = 2$ and $\mu = -\frac{1}{3}$.) So neither vector is a multiple of the other, and $\mathbf{a}$ and $\mathbf{b}$ are $\boxed{\text{independent}}$.

Check. $\|\hat{\mathbf{v}}\| = \frac{1}{7}\sqrt{4 + 36 + 9} = 1$. Recombining, $3\mathbf{a} + 2\mathbf{b} = (6, 3) + (2, -6) = (8, -3) = \mathbf{c}$. This check also exercises step 2.

Pitfalls

- Subtracting the wrong way round. $P - Q = (-2, 6, -3)$ has the same length, 7, so the distance survives, but the direction and the unit vector are reversed.
- Dividing by the wrong number. The component sum here is $2 - 6 + 3 = -1$, and dividing by it gives $(-2, 6, -3)$, of length 7. Dividing by $\left\|\overrightarrow{PQ}\right\|^2 = 49$ gives a vector of length $\frac{1}{7}$. The check $\|\hat{\mathbf{v}}\| = 1$ catches both.
- Taking "no two are parallel" to mean independent. No two of $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are parallel, yet $\mathbf{c} = 3\mathbf{a} + 2\mathbf{b}$ is a combination of the other two. So the three together are not independent; in $\mathbb{R}^2$ three vectors never are.

In the notes. Unit 4: Components, Unit 4: Addition, subtraction and scalar multiplication, Unit 4: Magnitude and unit vectors, and Unit 4: Linear combinations, independence and basis (intuition).

Practise it. the Review set on this method

4.2 The angle between vectors, and between planes

Problem. Find the angle between two vectors in $\mathbb{R}^2$, $\mathbb{R}^3$ or $\mathbb{R}^n$, where its cosine is the cosine similarity; find the angle between two planes.

Method

1. Collect the ingredients: $\mathbf{u} \cdot \mathbf{v}$, $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$. If the vectors join points, form them first by subtracting coordinates, $\overrightarrow{PQ} = (q_1 - p_1,\ q_2 - p_2,\ q_3 - p_3)$.
2. Divide: $\cos\theta = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\ \|\mathbf{v}\|}$, simplified. The sign of $\mathbf{u} \cdot \mathbf{v}$ alone gives the type of angle: positive acute, negative obtuse, zero a right angle, except that parallel vectors give $\theta = 0$ (same way) or π (opposite ways).
3. Give θ in $[0, \pi]$: exactly if $\cos\theta$ is a standard value, otherwise the exact $\cos\theta$ and then $\theta \approx \ldots^\circ$.
4. Two planes: read the normals off the coefficients. The plane $ax + by + cz = d$ has normal (a, b, c).
5. Take steps 1–3 on the normals, and report the angle between the planes in $[0, \frac{\pi}{2}]$, the smaller of θ and $\pi - \theta$, from $|\cos\theta|$. Perpendicular planes give $\frac{\pi}{2}$.

Variants. Orthogonality is the case $\mathbf{u} \cdot \mathbf{v} = 0$. A *proof* from the dot-product rules writes each squared length as a dot product, $\|\mathbf{w}\|^2 = \mathbf{w} \cdot \mathbf{w}$, expands, naming each rule it uses (distributive, scalar, commutative), and simplifies to the claim.

Example. Find the angle between $\mathbf{u} = (1, -2, 1)$ and $\mathbf{v} = (-2, 1, 1)$, and the angle between the planes $x + 2y - 2z = 1$ and $-4x + 3z = 2$.

The ingredients are $\mathbf{u} \cdot \mathbf{v} = -2 - 2 + 1 = -3$, $\|\mathbf{u}\| = \sqrt{1+4+1} = \sqrt{6}$ and $\|\mathbf{v}\| = \sqrt{4+1+1} = \sqrt{6}$. So

$$\cos\theta = \frac{-3}{\sqrt{6}\,\sqrt{6}} = -\frac{1}{2},$$

a standard value, and $\boxed{\theta = \frac{2\pi}{3} = 120^\circ}$: obtuse, as the negative dot product said.

The normals are $\mathbf{n}_1 = (1, 2, -2)$ and $\mathbf{n}_2 = (-4, 0, 3)$, with $\mathbf{n}_1 \cdot \mathbf{n}_2 = -4 + 0 - 6 = -10$, $\|\mathbf{n}_1\| = 3$ and $\|\mathbf{n}_2\| = 5$. So

$$\cos\theta = \frac{-10}{3 \cdot 5} = -\frac{2}{3},$$

not a standard value. The angle $\varphi \in [0, \frac{\pi}{2}]$ between the planes has $\cos\varphi = |\cos\theta| = \frac{2}{3}$:

$$\boxed{\cos\varphi = \tfrac{2}{3}}, \qquad \boxed{\varphi \approx 48.2^\circ}$$

Check. The law of cosines, from which the formula came: $\mathbf{u} - \mathbf{v} = (3, -3, 0)$ has $\|\mathbf{u} - \mathbf{v}\|^2 = 18$, and $\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\,\|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta = 6 + 6 + 6 = 18$. For the planes, the reversed normal $(4, 0, -3)$ describes the same second plane, as $4x - 3z = -2$, and gives $\cos\varphi = \frac{10}{15} = \frac{2}{3}$ directly.

Pitfalls

- Stopping at the dot product. $\mathbf{u} \cdot \mathbf{v} = -3$ is not the cosine, which always lies in $[-1, 1]$: divide by the lengths.
- Reporting 131.8° for the planes. That is the angle between the chosen normals; the planes meet at the angle in $[0, \frac{\pi}{2}]$, from $|\cos\theta|$.
- Building a normal from the right-hand side. Only the coefficients of x, y and z make it, and a missing variable gives a 0, as in $(-4, 0, 3)$.

In the notes. Unit 4: The scalar (dot) product, Unit 4: Beyond three dimensions: $\mathbb{R}^n$ and a data-science note, for cosine similarity, and Unit 4: Where a line meets a plane; the angle between two planes.

Practise it. the Review set on this method

4.3 Projection, resolving, work

Problem. Project one vector onto another; resolve a vector into a part along a direction and a part across it; find the work done by a force and its component along the motion.

Method

1. Compute $\mathbf{u} \cdot \mathbf{v}$ and $\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$.
2. The projection of $\mathbf{u}$ onto $\mathbf{v}$ is the multiple of $\mathbf{v}$

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}\,\mathbf{v}.$$

3. To resolve $\mathbf{u}$: the parallel part is $\mathbf{u}_\parallel = \operatorname{proj}_{\mathbf{v}} \mathbf{u}$, and the perpendicular part is the remainder, $\mathbf{u}_\perp = \mathbf{u} - \mathbf{u}_\parallel$.
4. Check that $\mathbf{u}_\perp \cdot \mathbf{v} = 0$ and $\mathbf{u}_\parallel + \mathbf{u}_\perp = \mathbf{u}$.

5. The scalar projection, the signed length of the shadow, is $\dfrac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|} = \mathbf{u}\cdot\hat{\mathbf{v}}$.
6. Work: a constant force $\mathbf{F}$ moving through a displacement $\mathbf{d}$ does work $W = \mathbf{F}\cdot\mathbf{d}$. Its scalar component along the motion is $\mathbf{F}\cdot\hat{\mathbf{d}} = W/\|\mathbf{d}\|$.

Example. A force $\mathbf{F} = (5, 2, 4)$ (newtons) moves its point of application through $\mathbf{d} = (2, 2, 1)$ (metres). Find the work done and the scalar component of $\mathbf{F}$ along the motion, and resolve $\mathbf{F}$ into parts along and across the motion.

$W = \mathbf{F}\cdot\mathbf{d} = 10 + 4 + 4$, so $\boxed{W = 18\ \text{J}}$. With $\|\mathbf{d}\| = \sqrt{4+4+1} = 3$, the scalar component along the motion is $W/\|\mathbf{d}\| = \boxed{6\ \text{N}}$.

For the parts, $\mathbf{F}\cdot\mathbf{d} = 18$ and $\|\mathbf{d}\|^2 = 9$, so $\mathbf{F}_{\parallel} = \frac{18}{9}(2, 2, 1) = (4, 4, 2)$, and the remainder is $\mathbf{F}_{\perp} = (5, 2, 4) - (4, 4, 2)$:

$$\boxed{\mathbf{F}_{\parallel} = (4, 4, 2)\ \text{N}}, \qquad \boxed{\mathbf{F}_{\perp} = (1, -2, 2)\ \text{N}}$$

Check. $\mathbf{F}_{\perp}\cdot\mathbf{d} = 2 - 4 + 2 = 0$ and $\mathbf{F}_{\parallel} + \mathbf{F}_{\perp} = (5, 2, 4) = \mathbf{F}$. The parallel part has length $\|(4, 4, 2)\| = 6$, the scalar component, and Pythagoras holds: $\|\mathbf{F}\|^2 = 45 = 36 + 9$.

Pitfalls

- Dividing by $\|\mathbf{d}\|$ instead of $\|\mathbf{d}\|^2$. It gives $6(2, 2, 1) = (12, 12, 6)$, three times too long. The vector projection divides by the length squared, the scalar projection by the length.
- Projecting the wrong way round. $\text{proj}_{\mathbf{d}}\,\mathbf{F}$ lies along $\mathbf{d}$; $\text{proj}_{\mathbf{F}}\,\mathbf{d}$ lies along $\mathbf{F}$, and is a different vector.
- Mixing up the two components. The scalar component is the number 6 N, the parallel part the vector $(4, 4, 2)$ N; and work is a scalar, in joules.

In the notes. Unit 4: The scalar (dot) product, with its projection example and the remark on work.

Practise it. the Review set on this method

4.4 The cross product: normal, area, volume, torque

Problem. In $\mathbb{R}^3$, find a vector perpendicular to two given vectors, the area of a parallelogram or a triangle, the volume of a parallelepiped, or a torque.

Method

1. Compute $\mathbf{u}\times\mathbf{v}$ from its components $u_2v_3 - u_3v_2$, $u_3v_1 - u_1v_3$ and $u_1v_2 - u_2v_1$, showing every product. The determinant with $\mathbf{i}, \mathbf{j}, \mathbf{k}$ in the top row is a way to remember them.
2. Check: the result dotted with $\mathbf{u}$, and with $\mathbf{v}$, gives 0.
3. Normal: $\mathbf{u}\times\mathbf{v}$ is perpendicular to both factors. Divide out any common factor; for a unit normal, divide by its length.
4. Area: $\|\mathbf{u}\times\mathbf{v}\|$ for the parallelogram on $\mathbf{u}$ and $\mathbf{v}$, and half of it for the triangle. For a triangle ABC, take $\mathbf{u} = \overrightarrow{AB}$ and $\mathbf{v} = \overrightarrow{AC}$.
5. Volume (optional enrichment, as in the notes): compute $\mathbf{v}\times\mathbf{w}$ first, then $\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})$. The volume of the parallelepiped is its absolute value, and 0 means the three vectors are coplanar.
6. Torque: $\boldsymbol{\tau} = \mathbf{r}\times\mathbf{F}$, by step 1, with $\mathbf{r}$ the position of the point where $\mathbf{F}$ acts, measured from

the pivot.

Example. Let $A(0,1,1)$, $B(2,4,-1)$, $C(1,3,-1)$ and $D(1,2,4)$. (a) Find a unit vector perpendicular to the plane of A, B and C, and the area of the triangle ABC. (b) Find the volume of the parallelepiped on $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$. (c) Find the torque about a pivot of the force $\mathbf{F} = (3,0,-1)$ (newtons) acting at $\mathbf{r} = (0,2,1)$ (metres) from it.

(a) $\overrightarrow{AB} = (2,3,-2)$ and $\overrightarrow{AC} = (1,2,-2)$. The components of their cross product are $3\cdot(-2) - (-2)\cdot 2 = -2$, $(-2)\cdot 1 - 2\cdot(-2) = 2$ and $2\cdot 2 - 3\cdot 1 = 1$:

$$\boxed{\overrightarrow{AB} \times \overrightarrow{AC} = (-2,2,1)}$$

Check: $\overrightarrow{AB}\cdot(-2,2,1) = -4+6-2 = 0$ and $\overrightarrow{AC}\cdot(-2,2,1) = -2+4-2 = 0$. Its length is $\sqrt{4+4+1} = 3$, so

$$\boxed{\hat{\mathbf{n}} = \pm\tfrac{1}{3}(-2,2,1)}, \qquad \boxed{\text{area} = \tfrac{3}{2}}$$

(b) The box on $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$ is the same whichever is called $\mathbf{u}$, so take $\mathbf{u} = \overrightarrow{AD} = (1,1,3)$ and reuse the cross product: $\overrightarrow{AD}\cdot(\overrightarrow{AB}\times\overrightarrow{AC}) = -2+2+3 = 3$, so the volume is $\boxed{3}$, and the four points are not coplanar.

(c) $\boldsymbol{\tau} = \mathbf{r}\times\mathbf{F}$ has components $2\cdot(-1) - 1\cdot 0 = -2$, $1\cdot 3 - 0\cdot(-1) = 3$ and $0\cdot 0 - 2\cdot 3 = -6$:

$$\boxed{\boldsymbol{\tau} = (-2,3,-6)\ \text{N m}}$$

of magnitude 7 N m. Check: $\boldsymbol{\tau}\cdot\mathbf{r} = 0+6-6 = 0$ and $\boldsymbol{\tau}\cdot\mathbf{F} = -6+0+6 = 0$.

Check. The area also follows from $\|\mathbf{u}\|\,\|\mathbf{v}\|\sin\theta$. Here $\left\|\overrightarrow{AB}\right\| = \sqrt{17}$, $\left\|\overrightarrow{AC}\right\| = 3$ and $\overrightarrow{AB}\cdot\overrightarrow{AC} = 12$, so $\cos\theta = \frac{4}{\sqrt{17}}$ and $\sin\theta = \frac{1}{\sqrt{17}}$. The parallelogram's area is $\sqrt{17}\cdot 3\cdot\frac{1}{\sqrt{17}} = 3$, twice the triangle's.

Pitfalls

- Getting the middle component's sign wrong. It is $u_3v_1 - u_1v_3$, the minus sign on $\mathbf{j}$ in the determinant: above, $(-2)\cdot 1 - 2\cdot(-2) = 2$, not -2.
- Forgetting the half: 3 is the area of the parallelogram, and the triangle's is $\frac{3}{2}$.
- Reversing the order. $\overrightarrow{AC}\times\overrightarrow{AB} = (2,-2,-1)$ is still a normal, but $\mathbf{F}\times\mathbf{r}$ is minus the torque. A negative triple product still gives a positive volume: take the absolute value.

In the notes. Unit 4: The vector (cross) product, and Unit 4: Optional: the scalar triple product.
Practise it. the Review set on this method

4.5 A line through two points; is a point on it?

Problem. Find the line through two points in vector, parametric or symmetric form, and decide whether a given point lies on it.

Method

1. The direction is $\mathbf{d} = \overrightarrow{AB}$: subtract the coordinates of A from those of B. A tidier multiple serves as well. The length $\left\|\overrightarrow{AB}\right\|$ is the distance from A to B.
2. Take A as the base point, with position vector $\mathbf{r}_0$. The vector form is $\mathbf{r} = \mathbf{r}_0 + t\,\mathbf{d}$, and in

coordinates the parametric form is $x = x_0 + t\,d_1$, $y = y_0 + t\,d_2$, $z = z_0 + t\,d_3$.

3. If asked, eliminate t for the symmetric form $\dfrac{x - x_0}{d_1} = \dfrac{y - y_0}{d_2} = \dfrac{z - z_0}{d_3}$, possible when no d_i is 0.
4. Check: some value of t returns B.
5. Point test: solve for t from one coordinate whose $d_i \neq 0$, then put that t into the others. The point is on the line only if every coordinate matches.

Example. Find the line through $A(3, 1, -2)$ and $B(-1, 3, 4)$ in vector, parametric and symmetric form, and the distance AB. Do $C(7, -1, -8)$ and $D(1, 2, 2)$ lie on it?

$\overrightarrow{AB} = (-4, 2, 6) = 2(-2, 1, 3)$, so $AB = \sqrt{16 + 4 + 36} = \sqrt{56} = \boxed{2\sqrt{14}}$, and the tidier direction is $\mathbf{d} = (-2, 1, 3)$. With base point A,

$$\boxed{\mathbf{r} = (3, 1, -2) + t(-2, 1, 3)}$$

that is, $x = 3 - 2t$, $y = 1 + t$, $z = -2 + 3t$. Eliminating t,

$$\boxed{\frac{x - 3}{-2} = y - 1 = \frac{z + 2}{3}}$$

Putting $t = 2$ gives $(3 - 4,\ 1 + 2,\ -2 + 6) = (-1, 3, 4) = B$.

For C: $3 - 2t = 7$ gives $t = -2$, and then $y = 1 - 2 = -1$ and $z = -2 - 6 = -8$ both match, so $\boxed{C \text{ is on the line}}$, at $t = -2$. For D: $3 - 2t = 1$ gives $t = 1$; then $y = 2$ matches, but $z = -2 + 3 = 1 \neq 2$, so $\boxed{D \text{ is not on the line}}$.

Check. $\overrightarrow{AC} = (4, -2, -6) = -2\mathbf{d}$ is parallel to the direction, as it must be for a point of the line; $\overrightarrow{AD} = (-2, 1, 4)$ is not a multiple of $\mathbf{d}$.

Pitfalls

- Testing too few coordinates. D matches in x and in y, and fails only in z.
- Using $\overrightarrow{AB}$ as the base point. $\mathbf{r} = (-4, 2, 6) + t(-2, 1, 3)$ is a parallel line that misses A and B.
- Writing a symmetric form with a 0 in a denominator. When some $d_i = 0$, that coordinate is constant along the line; state it separately.

In the notes. Unit 4: Components, for the vector between two points, and Unit 4: The equation of a line.

Practise it. the Review set on this method

4.6 A plane through three points

Problem. Find the plane through three points, in point-normal, Cartesian or parametric form.

Method

1. *Two vectors in the plane.* With P as the base point, form $\overrightarrow{PQ}$ and $\overrightarrow{PR}$.
2. *A normal, by the cross product*: $\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR}$ (Method 4.4). Any nonzero multiple serves, so divide out a common factor.

3. *The equation.* Write the point-normal form $\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$ at P, that is $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$, and multiply it out to $ax + by + cz = d$, with $d = \mathbf{n} \cdot \mathbf{r}_0$.
4. Check: all three points satisfy it.
5. Parametric form, if asked: $\mathbf{r} = \mathbf{r}_0 + s\,\overrightarrow{PQ} + t\,\overrightarrow{PR}$, with $s, t \in \mathbb{R}$.

Why it works. A point $\mathbf{r}$ lies in the plane exactly when $\mathbf{r} - \mathbf{r}_0$ lies flat in it, that is, when it is orthogonal to $\mathbf{n}$. If $\overrightarrow{PQ} \times \overrightarrow{PR} = \mathbf{0}$, the three points are on one line and do not fix a plane.

Example. Find the plane through $P(1, -1, 2)$, $Q(-1, 0, 1)$ and $R(1, 2, 3)$ in Cartesian and in parametric form.

Two vectors in the plane: $\overrightarrow{PQ} = (-2, 1, -1)$ and $\overrightarrow{PR} = (0, 3, 1)$.

A normal: the components of $\overrightarrow{PQ} \times \overrightarrow{PR}$ are $1 \cdot 1 - (-1) \cdot 3 = 4$, $(-1) \cdot 0 - (-2) \cdot 1 = 2$ and $(-2) \cdot 3 - 1 \cdot 0 = -6$. So it is $(4, 2, -6) = 2(2, 1, -3)$; take $\mathbf{n} = (2, 1, -3)$.

The equation: point-normal form at P,

$$2(x - 1) + 1(y + 1) - 3(z - 2) = 0,$$

which multiplies out to $2x + y - 3z + 5 = 0$:

$$\boxed{2x + y - 3z = -5}$$

Check: P gives $2 - 1 - 6 = -5$, Q gives $-2 + 0 - 3 = -5$ and R gives $2 + 2 - 9 = -5$.

In parametric form,

$$\mathbf{r} = (1, -1, 2) + s(-2, 1, -1) + t(0, 3, 1),$$

that is, $\boxed{\mathbf{r} = (1 - 2s,\ -1 + s + 3t,\ 2 - s + t)}$.

Check. The normal is perpendicular to both vectors in the plane: $(2, 1, -3) \cdot (-2, 1, -1) = -4 + 1 + 3 = 0$ and $(2, 1, -3) \cdot (0, 3, 1) = 0 + 3 - 3 = 0$. The parametric point at $s = 1$, $t = -1$ is $(-1, -3, 0)$, and $2(-1) + (-3) - 3 \cdot 0 = -5$.

Pitfalls

- Crossing the position vectors instead of the vectors in the plane: $\overrightarrow{OQ} \times \overrightarrow{OR} = (-2, 4, -2)$ is not perpendicular to $\overrightarrow{PQ}$ (their dot product is 10).
- Stopping at $2x + y - 3z = 0$. The right-hand side is $d = \mathbf{n} \cdot \mathbf{r}_0 = -5$; it is 0 only for a plane through the origin.
- Checking only the base point. P satisfies the equation by construction; Q and R are what test the normal.

In the notes. Unit 4: The equation of a plane, with its worked example.

Practise it. the Review set on this method

4.7 Where a line meets a plane; distance from a point to a plane

Problem. Find where a line meets a plane, or show that it misses the plane or lies in it; find the foot of the perpendicular from a point to a plane, and the distance.

Method

1. Write the coordinates of a point of the line in terms of t: $x = x_0 + t\,d_1$, $y = y_0 + t\,d_2$, $z = z_0 + t\,d_3$.
2. Substitute them into the plane's equation, and solve for t.
3. Put that t back into the line: it gives the crossing point. Check that the point satisfies the plane's equation.
4. If t cancels, then $\mathbf{d}\cdot\mathbf{n} = 0$ and the direction lies flat. The base point decides: on the plane, the line lies in it (every t works); off it, the line is parallel (no t works).
5. Distance from a point $\mathbf{p}$: step from $\mathbf{p}$ along the normal, $\mathbf{p} + t\,\mathbf{n}$. Substitute into the plane's equation and solve for t; that point is the foot of the perpendicular.
6. The distance is $|t|\ \|\mathbf{n}\|$. Check it by the formula $\dfrac{|ap_1 + bp_2 + cp_3 - d|}{\|\mathbf{n}\|}$.

Example. Take the plane $2x - 3y + 6z = 5$. (a) Where does the line $\mathbf{r} = (3,1,0) + t(1,1,0)$ meet it? (b) And the line $\mathbf{r} = (1,1,1) + t(3,0,-1)$? (c) Find the foot of the perpendicular from $P(2,-1,2)$ to the plane, and the distance from P to it.

(a) A point of the line is $(3+t,\ 1+t,\ 0)$. Substituting,

$$2(3+t) - 3(1+t) + 6\cdot 0 = 5 \iff 3 - t = 5,$$

so $t = -2$, and the crossing point is $\boxed{(1,-1,0)}$. It satisfies $2 + 3 + 0 = 5$.

(b) A point of the line is $(1+3t,\ 1,\ 1-t)$. Substituting,

$$2(1+3t) - 3 + 6(1-t) = 5 \iff 5 = 5,$$

which holds for every t. The t cancelled because $\mathbf{d}\cdot\mathbf{n} = 6 + 0 - 6 = 0$, and the base point is on the plane $(2 - 3 + 6 = 5)$, so $\boxed{\text{the line lies in the plane}}$.

(c) The normal is $\mathbf{n} = (2,-3,6)$, with $\|\mathbf{n}\| = \sqrt{4+9+36} = 7$. Step from P: $\mathbf{p} + t\,\mathbf{n} = (2+2t,\ -1-3t,\ 2+6t)$ lies on the plane when

$$2(2+2t) - 3(-1-3t) + 6(2+6t) = 5,$$

that is $19 + 49t = 5$, so $t = -\frac{14}{49} = -\frac{2}{7}$. The foot is $(2,-1,2) - \frac{2}{7}(2,-3,6)$:

$$\boxed{F = \left(\tfrac{10}{7}, -\tfrac{1}{7}, \tfrac{2}{7}\right)}$$

and the distance is $|t|\ \|\mathbf{n}\| = \frac{2}{7}\cdot 7 = \boxed{2}$.

Check. The formula gives $\dfrac{|2\cdot 2 - 3\cdot(-1) + 6\cdot 2 - 5|}{7} = \dfrac{14}{7} = 2$, and the foot is on the plane: $\frac{20}{7} + \frac{3}{7} + \frac{12}{7} = 5$. The foot is also 2 from P: $\overrightarrow{FP} = \frac{1}{7}(4,-6,12)$ has length $\frac{1}{7}\sqrt{196} = 2$.

Pitfalls

- Stopping at t. In (a), $t = -2$ is a parameter value, not the point: put it back into the line.
- Calling the line in (b) parallel because $\mathbf{d}\cdot\mathbf{n} = 0$. That says only that the direction lies flat; the base point decides between parallel and contained.
- Giving $|t| = \frac{2}{7}$ as the distance. The step was $t\,\mathbf{n}$, so the distance is $|t|\ \|\mathbf{n}\| = 2$, unless $\mathbf{n}$ is a unit vector.

In the notes. Unit 4: Where a line meets a plane; the angle between two planes, with its note on when a line crosses a plane and its distance example.

Practise it. the Review set on this method

Unit 5 Matrices

Five methods, and each starts from the sizes and the shape of the matrix. A short proof about matrices (a property of the inverse or of the transpose) follows one pattern: state the goal, rewrite it with the named rules (associativity, the transpose rules, uniqueness of the inverse) and end with a concluding sentence. For a trace identity, write out the entries of both sides instead.

Which method?

The question asks for	Method
whether a sum or a product is defined, and its size	5.1, compare the sizes first
a product, a transpose or a trace	5.1, row times column
a 3×3 determinant with zeros in it	5.2, expand along the row or column with most zeros
the value of a parameter that makes a matrix singular	5.2, solve det = 0
the inverse of a 2×2 matrix	5.3, swap, negate, divide by det
the inverse of a 3×3 matrix that is not triangular	5.3, cofactors and the adjugate
the inverse of a triangular or larger matrix	5.3, row-reduce $(\,\mathbf{A} \mid \mathbf{I}\,)$
the solution of a system by the inverse method, or with $\mathbf{A}^{-1}$ known	5.4, variant: $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ ($\mathbf{A}^{-1}$ by 5.3)
the solution of a system, otherwise	5.4, Gaussian elimination
whether a system has one, no or infinitely many solutions	5.4, read the echelon form, $0 = c$ first
the matrix of a linear map of the plane, or of two maps in turn	5.5, columns are the images of $\mathbf{i}, \mathbf{j}$
the effect of a map on area and orientation	5.5, the sign and size of det

5.1 Products, transpose and trace: check the sizes first

Problem. Decide whether a sum or a product of matrices is defined, and compute it; find a transpose or a trace.

Method

1. Write down the size of every matrix, $m \times n$ (m rows, n columns).
2. Sum or difference: the sizes must be equal. Then, as for a scalar multiple, work entry by entry.
3. Product $\mathbf{AB}$: write the sizes as $(m \times n)(n' \times p)$. If $n \neq n'$ the product is undefined: stop. Otherwise it is $m \times p$.
4. The (i, j) entry of $\mathbf{AB}$ is row i of $\mathbf{A}$ dotted with column j of $\mathbf{B}$.
5. Transpose: swap rows and columns, so the (i, j) entry of $\mathbf{A}^{\mathsf{T}}$ is a_{ji}. For a product, $(\mathbf{AB})^{\mathsf{T}} = \mathbf{B}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}$.
6. Trace, of a square matrix only: the sum of the diagonal entries.

Example. Let $\mathbf{A} = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 3 & -2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 4 \\ 0 & -1 \\ 3 & 2 \end{pmatrix}$. Which of $\mathbf{A}+\mathbf{B}$, $\mathbf{A}+\mathbf{B}^{\mathsf{T}}$, $\mathbf{AB}$ and $\mathbf{BA}$ are defined? Compute $\mathbf{A}+\mathbf{B}^{\mathsf{T}}$, $\mathbf{AB}$ and $\operatorname{tr}(\mathbf{AB})$.

$\mathbf{A}$ is 2×3 and $\mathbf{B}$ is 3×2, so $\boxed{\mathbf{A}+\mathbf{B} \text{ is undefined}}$. $\mathbf{B}^{\mathsf{T}}$ is 2×3, so $\mathbf{A}+\mathbf{B}^{\mathsf{T}}$ is defined and 2×3. The product $(2\times 3)(3\times 2)$ has matching inner sizes and is 2×2; $\mathbf{BA}$ is $(3\times 2)(2\times 3)$, so it is 3×3. Entrywise,

$$\mathbf{A}+\mathbf{B}^{\mathsf{T}} = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 3 & -2 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 3 \\ 4 & -1 & 2 \end{pmatrix} = \boxed{\begin{pmatrix} 3 & -1 & 3 \\ 5 & 2 & 0 \end{pmatrix}}.$$

Row times column,

$$\mathbf{AB} = \begin{pmatrix} 2+0+0 & 8+1+0 \\ 1+0-6 & 4-3-4 \end{pmatrix} = \boxed{\begin{pmatrix} 2 & 9 \\ -5 & -3 \end{pmatrix}},$$

and the trace is the diagonal sum, $2+(-3)$: $\boxed{\operatorname{tr}(\mathbf{AB}) = -1}$.

Check. The cyclic property says $\operatorname{tr}(\mathbf{BA}) = \operatorname{tr}(\mathbf{AB})$, although $\mathbf{BA}$ is 3×3. Its diagonal entries are $1\cdot 2+4\cdot 1 = 6$, $0\cdot(-1)+(-1)\cdot 3 = -3$ and $3\cdot 0+2\cdot(-2) = -4$, and $6-3-4 = -1$. That tests only the diagonal. For every entry use associativity, $(\mathbf{AB})\mathbf{v} = \mathbf{A}(\mathbf{Bv})$, with $\mathbf{v} = (1,1)^{\mathsf{T}}$: $\mathbf{Bv} = (5,-1,5)^{\mathsf{T}}$ and $\mathbf{A}(\mathbf{Bv}) = (11,-8)^{\mathsf{T}}$, the row sums $2+9$ and $-5-3$ of $\mathbf{AB}$. A wrong entry changes a row sum.

Pitfalls

- Adding $\mathbf{A}$ and $\mathbf{B}$ because both have six entries. The sizes 2×3 and 3×2 differ, so the sum does not exist.
- Multiplying entry by entry, or giving a product the inner sizes: the size of $\mathbf{BA}$ comes from the outer ones, 3×3.
- Transposing a product without reversing it. Here $\mathbf{A}^{\mathsf{T}}\mathbf{B}^{\mathsf{T}}$ is 3×3, so it cannot be $(\mathbf{AB})^{\mathsf{T}}$, which is 2×2.

In the notes. Unit 5: Addition, subtraction and scalar multiplication, Unit 5: Matrix multiplication, and Unit 5: Transpose and trace.

Practise it. the Review set on this method

5.2 Determinants by cofactor expansion; the parameter that makes a matrix singular

Problem. Evaluate the determinant of a 2×2 or 3×3 matrix; decide whether the matrix is invertible; find the values of a parameter for which it is singular.

Method

1. For a 2×2 matrix, $\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad-bc$.
2. For a 3×3 matrix, choose the row or column with the most zeros (the first row if none stands out).
3. Write each entry of that row or column times its cofactor $C_{ij} = (-1)^{i+j}M_{ij}$. The sign follows the checkerboard $\begin{smallmatrix} + & - & + \\ - & + & - \\ + & - & + \end{smallmatrix}$, and the minor M_{ij} is the 2×2 determinant left when row i and column j are deleted. A zero entry drops out.

4. Evaluate the 2×2 minors and add up.
5. The matrix is invertible exactly when $\det \neq 0$; $\det = 0$ means singular.
6. For a parameter k: find det as an expression in k, set it equal to 0 and solve. Those values, and only those, make the matrix singular.

Variants. A *triangular* matrix has determinant equal to the product of its diagonal entries, with no expansion needed. A larger matrix expands the same way: for a 4×4 the minors are 3×3 determinants.

Example. (a) Evaluate $\det \mathbf{A}$ for $\mathbf{A} = \begin{pmatrix} 2 & 1 & -3 \\ 4 & 0 & 5 \\ -1 & 0 & 2 \end{pmatrix}$. (b) For which k is $\mathbf{B} = \begin{pmatrix} k & 1 & 0 \\ 3 & k & 2 \\ 0 & 1 & 1 \end{pmatrix}$ singular?

(a) Column 2 has two zeros, so expand along it. Only the entry $a_{12} = 1$ survives, and position $(1, 2)$ carries the sign $-$:

$$\det \mathbf{A} = -1 \begin{vmatrix} 4 & 5 \\ -1 & 2 \end{vmatrix} = -(8 + 5) = \boxed{-13}.$$

(b) Expand along the first row, whose third entry is 0:

$$\det \mathbf{B} = k \begin{vmatrix} k & 2 \\ 1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 3 & 2 \\ 0 & 1 \end{vmatrix} = k(k - 2) - 3.$$

So $\det \mathbf{B} = k^2 - 2k - 3 = (k - 3)(k + 1)$, which is 0 for $\boxed{k = -1 \text{ or } k = 3}$.

Check. (a) Expanding along the first row instead gives $2 \cdot 0 - 1 \cdot 13 + (-3) \cdot 0 = -13$. (b) At $k = 3$ the rows of $\mathbf{B}$ are $(3, 1, 0)$, $(3, 3, 2)$, $(0, 1, 1)$, and the second is the first plus twice the third. At $k = -1$ they are $(-1, 1, 0)$, $(3, -1, 2)$, $(0, 1, 1)$, and the second is twice the third minus three times the first. Either way the three rows lie in one plane, so the determinant must be 0: both roots check.

Pitfalls

- Dropping the checkerboard sign. In (a) the cofactor of a_{12} is -13, so $\det \mathbf{A} = -13$, not 13.
- Using the diagonal product on a matrix that is not triangular.
- Losing a root: $k^2 - 2k - 3 = 0$ has two solutions, and both make $\mathbf{B}$ singular.

In the notes. Unit 5: The determinant, and Unit 5: The inverse, for invertible exactly when $\det \neq 0$.

Practise it. the Review set on this method

5.3 The inverse: the 2×2 formula, cofactors, or row reduction

Problem. Find the inverse of a square matrix, or show that it has none.

Method

1. Find $\det \mathbf{A}$ (Method 5.2). If it is 0 there is no inverse: stop.
2. A 2×2 matrix: swap the diagonal entries, negate the off-diagonal ones, and divide by the determinant, $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \dfrac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

3. A 3×3 matrix by cofactors: find all nine cofactors $C_{ij} = (-1)^{i+j} M_{ij}$ and write the cofactor matrix $\mathbf{C} = (C_{ij})$.
4. Transpose it to get the adjugate, $\operatorname{adj}(\mathbf{A}) = \mathbf{C}^{\mathsf{T}}$, and divide by the determinant: $\mathbf{A}^{-1} = \dfrac{1}{\det \mathbf{A}} \operatorname{adj}(\mathbf{A})$.
5. Or by row reduction: write $(\,\mathbf{A} \mid \mathbf{I}\,)$ and apply row operations until the left block is $\mathbf{I}$ (echelon form, then scale each leading entry to 1 and clear above it). The right block is then $\mathbf{A}^{-1}$.
6. Check one product: $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$.

Example. Invert (a) $\mathbf{P} = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix}$ and (b) $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 3 & -1 \\ 2 & 1 & -1 \end{pmatrix}$.

(a) $\det \mathbf{P} = 3(-1) - (-2)(4) = 5 \neq 0$. Swap, negate and divide by 5:

$$\boxed{\mathbf{P}^{-1} = \frac{1}{5} \begin{pmatrix} -1 & 2 \\ -4 & 3 \end{pmatrix}}$$

(b) Expand along the first row: $\det \mathbf{A} = 1 \cdot \begin{vmatrix} 3 & -1 \\ 1 & -1 \end{vmatrix} = -2$. Deleting each row and column in turn and attaching the checkerboard sign gives

$$\mathbf{C} = \begin{pmatrix} -2 & 1 & -3 \\ 0 & -1 & -1 \\ 0 & 1 & 3 \end{pmatrix},$$

for instance $C_{12} = -\begin{vmatrix} 3 & -1 \\ 2 & -1 \end{vmatrix} = 1$ and $C_{23} = -\begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = -1$. Transpose, then divide by -2:

$$\operatorname{adj}(\mathbf{A}) = \mathbf{C}^{\mathsf{T}} = \begin{pmatrix} -2 & 0 & 0 \\ 1 & -1 & 1 \\ -3 & -1 & 3 \end{pmatrix},$$

$$\boxed{\mathbf{A}^{-1} = \frac{1}{2} \begin{pmatrix} 2 & 0 & 0 \\ -1 & 1 & -1 \\ 3 & 1 & -3 \end{pmatrix}}$$

Check. $\mathbf{P} \begin{pmatrix} -1 & 2 \\ -4 & 3 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$, so $\mathbf{P}\mathbf{P}^{-1} = \mathbf{I}$. And

$$\mathbf{A} \begin{pmatrix} 2 & 0 & 0 \\ -1 & 1 & -1 \\ 3 & 1 & -3 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

so $\mathbf{A}\mathbf{A}^{-1} = \frac{1}{2} \cdot 2\mathbf{I} = \mathbf{I}$.

Pitfalls

- Forgetting the transpose. In (b) $C_{12} = 1$, but the $(1,2)$ entry of the adjugate is $C_{21} = 0$.
- Dividing by $|\det \mathbf{A}| = 2$ instead of by -2: every entry of $\mathbf{A}^{-1}$ changes sign.
- Inverting entry by entry. $\mathbf{P}^{-1}$ is not the matrix of reciprocals $\frac{1}{3}$, $-\frac{1}{2}$, $\frac{1}{4}$, -1.

In the notes. Unit 5: The inverse, and Unit 5: Gaussian elimination, for the inverse by row reduction.

Practise it. the Review set on this method

5.4 Gaussian elimination and reading the echelon form

Problem. Solve a linear system $\mathbf{Ax} = \mathbf{b}$, or decide whether it has one solution, none or infinitely many.

Method

1. Write the augmented matrix $(\,\mathbf{A} \mid \mathbf{b}\,)$.
2. Reduce it to echelon form with labelled row operations $R_i \to R_i + kR_j$, stacked when done at once. Keep the entries integers (a step such as $R_2 \to 2R_2 + 3R_1$ is allowed). If the entry that should be the next leading entry is 0, swap in a lower row where it is not.
3. Look *first* for a row $(\,0 \;\cdots\; 0 \mid c\,)$ with $c \neq 0$. It reads $0 = c$: there is no solution. Stop.
4. Otherwise count the leading entries. With one for each unknown the solution is unique: back-substitute, from the last row up.
5. With fewer leading entries than unknowns, an unknown whose column has no leading entry is free. Set it equal to t, back-substitute, and write the solution as $\mathbf{x} = \mathbf{p} + t\,\mathbf{d}$, $t \in \mathbb{R}$: a line. With two free unknowns, set them equal to s and t: the solution $\mathbf{x} = \mathbf{p} + s\,\mathbf{u} + t\,\mathbf{v}$ is a plane.
6. Check the answer in every original equation (for a line, at $t = 0$ and $t = 1$).

Variants. *Gauss–Jordan* continues from the echelon form: scale each leading entry to 1 and clear above it. When the solution is unique this reaches $(\,\mathbf{I} \mid \mathbf{x}\,)$, and the solution is read off. The *inverse method* solves $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$, with $\mathbf{A}^{-1}$ found by Method 5.3 if it is not given. With $\mathbf{P}^{-1}$ from Method 5.3, the system $3x - 2y = 4$, $4x - y = 7$ gives

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{5}\begin{pmatrix} -1 & 2 \\ -4 & 3 \end{pmatrix}\begin{pmatrix} 4 \\ 7 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} -4+14 \\ -16+21 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

For a square system, $\det \mathbf{A} \neq 0$ guarantees exactly one solution; $\det = 0$ leaves none or infinitely many, and only the echelon form says which.

Example. For which values of a and b does the system

$$x + 2y - z = 2, \qquad 2x + 5y + 3z = 7, \qquad 3x + 7y + az = b$$

have exactly one solution, none, or infinitely many? Solve it in the last case.

Reduce the augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 2 & 5 & 3 & 7 \\ 3 & 7 & a & b \end{array}\right)$$

$$\xrightarrow[R_3\to R_3-3R_1]{R_2\to R_2-2R_1}\left(\begin{array}{ccc|c}1&2&-1&2\\0&1&5&3\\0&1&a+3&b-6\end{array}\right)$$

$$\xrightarrow{R_3\to R_3-R_2}\left(\begin{array}{ccc|c}1&2&-1&2\\0&1&5&3\\0&0&a-2&b-9\end{array}\right).$$

If $a \neq 2$ there are three leading entries for three unknowns: $\boxed{a \neq 2:\ \text{exactly one solution}}$. If $a = 2$ the last row is $(\,0\ 0\ 0 \mid b-9\,)$, which reads $0 = b-9$. So $\boxed{a = 2,\ b \neq 9:\ \text{no solution}}$.

If $a = 2$ and $b = 9$ the last row is all zeros, and z is free. Set $z = t$. Row 2 gives $y = 3 - 5t$, and row 1 gives $x = 2 - 2(3-5t) + t = -4 + 11t$:

$$\boxed{\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}-4\\3\\0\end{pmatrix} + t\begin{pmatrix}11\\-5\\1\end{pmatrix},\ t \in \mathbb{R}}$$

a line on which the three planes meet.

Check. With $a = 2$, $b = 9$: $t = 0$ gives $(-4, 3, 0)$ and $t = 1$ gives $(7, -2, 1)$, and $7 - 4 - 1 = 2$, $14 - 10 + 3 = 7$, $21 - 14 + 2 = 9$ (the first point works the same way). Also $\det \mathbf{A} = a - 2$ by cofactors, so $a \neq 2$ is exactly $\det \mathbf{A} \neq 0$.

Pitfalls

- Reading a zero row of coefficients as "infinitely many" before looking at the right-hand side. With $a = 2$, $b = 10$ the row is $(\,0\ 0\ 0 \mid 1\,)$: no solution.
- Changing the coefficients and forgetting the right-hand column, or using a row in the same step as it is being changed.
- Deciding between none and infinitely many from $\det \mathbf{A} = 0$. The determinant only says the solution is not unique.

In the notes. Unit 5: The inverse-matrix method, Unit 5: Gaussian elimination, and Unit 5: What the solutions look like.

Practise it. the Review set on this method

5.5 A matrix as a transformation of the plane

Problem. Find the matrix of a linear map of the plane, or of two maps done in turn; find the image of a point or a shape; describe the effect on area and orientation.

Method

1. Only a linear map has a matrix: a translation moves the origin, and no 2×2 matrix does. The columns of the matrix are the images of $\mathbf{i} = (1, 0)$ and $\mathbf{j} = (0, 1)$. Or take it from the standard list: rotation by θ, $\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$; reflection in $y = x$; enlargement; stretch; shear.
2. "Do **S**, then **R**" is the single matrix **RS**: the map done first stands on the right.
3. Apply the matrix to the vector, or to each corner of a shape. The edges between corners go to the edges between their images.

4. Read the determinant: |det| is the factor by which areas scale, a negative det flips the orientation, and det = 0 collapses the plane onto a line, or onto the origin for the zero matrix (the map is not invertible).
5. To name a map, see where it sends **i** and **j**.

Example. Let $\mathbf{S} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, the shear $(x, y) \mapsto (x + 2y, y)$, and let $\mathbf{T}$ be the map that sends **i** to $(1, 1)$ and **j** to $(2, -1)$. (a) Find $\mathbf{T}$. (b) Find the matrix $\mathbf{M}$ of "first $\mathbf{S}$, then $\mathbf{T}$". (c) Find the image of the triangle with corners $(0, 0)$, $(3, 0)$, $(1, 2)$ under $\mathbf{M}$, and the effect on area and orientation.

(a) The images of **i** and **j** are the columns: $\boxed{\mathbf{T} = \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}}$.

(b) The shear is done first, so it stands on the right:

$$\mathbf{M} = \mathbf{TS} = \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \boxed{\begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix}}$$

(c) $\mathbf{M}$ fixes $(0, 0)$ and sends $(3, 0)$ to $(3, 3)$ and $(1, 2)$ to $(1+8,\ 1+2)$. The image is the triangle with corners $\boxed{(0, 0),\ (3, 3),\ (9, 3)}$. Since $\boxed{\det \mathbf{M} = 1 - 4 = -3}$, areas are multiplied by 3 and the orientation is flipped.

Check. Half the 2×2 determinant of the two edges from $(0, 0)$ gives the signed area. Before: $\frac{1}{2}(3 \cdot 2 - 0 \cdot 1) = 3$. After: $\frac{1}{2}(3 \cdot 3 - 3 \cdot 9) = -9$: the area is $9 = 3 \times 3$, and the sign has reversed.

Pitfalls

- Multiplying in the order of the words. $\mathbf{ST} = \begin{pmatrix} 3 & 0 \\ 1 & -1 \end{pmatrix}$ applies $\mathbf{T}$ first, and sends $(1, 2)$ to $(3, -1)$, not $(9, 3)$.
- Writing the images of **i** and **j** as the rows instead of the columns.
- Reading $\det \mathbf{M} = -3$ as a negative area. The area scales by $|\det| = 3$; the sign is the orientation.

In the notes. Unit 5: Matrices as transformations, and Unit 5: The determinant, for the determinant as a signed area.

Practise it. the Review set on this method

Unit 6 Eigensystems

Six methods, and the first is a step in all the others. True-or-false questions and short proofs are not a method: answer each with a one-line reason that cites a result from the notes, or with one concrete counterexample, and set a proof out as set-up, derivation, conclusion.

6.1 The eigensystem: 2×2, 3×3, complex eigenvalues

Problem. Find the eigenvalues of a square matrix and an eigenvector for each, real or complex.

Method

1. Write $\det(\mathbf{A} - \lambda\mathbf{I})$ and expand it: a 2×2 directly, a 3×3 along the row or column with the most zeros.
2. Factor the characteristic polynomial and name its roots $\lambda_1, \lambda_2, \ldots$: these are the eigenvalues. A real 2×2 with a negative discriminant gives a complex pair $a \pm b\,\mathrm{i}$.
3. For each eigenvalue in turn, write $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$. For a 2×2, one row is enough: read off how v_1 and v_2 are related and choose a simple $\mathbf{v}$.
4. For a 3×3, row-reduce $\mathbf{A} - \lambda\mathbf{I}$, or read the equations off directly when the zeros make them immediate. Set the free unknown to 1 and back-substitute.
5. For a complex pair, find the eigenvector of $a + b\,\mathrm{i}$ from row 1; the eigenvector of $a - b\,\mathrm{i}$ is its conjugate. Only if asked: $|\lambda|$ sets the growth and $\arg\lambda$ the turning, on average per step over many steps; exactly at every step only for $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$.
6. Check one pair with $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, and the eigenvalues with $\sum \lambda_i = \operatorname{tr}\mathbf{A}$ and $\prod \lambda_i = \det\mathbf{A}$.

Example. Find the eigenvalues and eigenvectors of (a) $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 4 & -2 \\ 1 & 1 & 1 \end{pmatrix}$ and (b) $\mathbf{B} = \begin{pmatrix} 4 & -5 \\ 2 & -2 \end{pmatrix}$.

(a) Row 1 has two zeros, so expand along it:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (1 - \lambda)\big[(4 - \lambda)(1 - \lambda) + 2\big] = (1 - \lambda)(\lambda - 2)(\lambda - 3),$$

since $(4 - \lambda)(1 - \lambda) + 2 = \lambda^2 - 5\lambda + 6$. So $\lambda_1 = 1$, $\lambda_2 = 2$, $\lambda_3 = 3$. For $\lambda_1 = 1$, swap R_1 and R_3 of $\mathbf{A} - \mathbf{I}$, then take $R_2 \to R_2 - R_1$:

$$\mathbf{A} - \mathbf{I} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 3 & -2 \\ 1 & 1 & 0 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & -2 \\ 0 & 0 & 0 \end{pmatrix}.$$

With $v_3 = 1$, back-substitution gives $v_2 = 1$ and $v_1 = -1$. For $\lambda_2 = 2$ and $\lambda_3 = 3$ the first row of $\mathbf{A} - \lambda\mathbf{I}$ is $(1 - \lambda, 0, 0)$, so $v_1 = 0$ at once. Then row 2 reads $2v_2 - 2v_3 = 0$ for $\lambda = 2$ and $v_2 - 2v_3 = 0$ for $\lambda = 3$:

$$\boxed{\lambda = 1 : \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}}, \qquad \boxed{\lambda = 2 : \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}},$$

$$\boxed{\lambda = 3 : \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}}$$

(b) $\det(\mathbf{B} - \lambda\mathbf{I}) = (4-\lambda)(-2-\lambda) + 10 = \lambda^2 - 2\lambda + 2 = (\lambda-1)^2 + 1$, so $\lambda = 1 \pm \mathrm{i}$. For $\lambda = 1+\mathrm{i}$, row 1 of $\mathbf{B} - \lambda\mathbf{I}$ reads $(3-\mathrm{i})v_1 - 5v_2 = 0$; take $v_1 = 5$, $v_2 = 3 - \mathrm{i}$. The eigenvector of $1 - \mathrm{i}$ is the conjugate:

$$\boxed{\lambda = 1 \pm \mathrm{i}, \quad \mathbf{v} = (5,\ 3 \mp \mathrm{i})^{\mathsf{T}}}$$

If the geometry is asked for: $|1+\mathrm{i}| = \sqrt{2}$ and $\arg(1+\mathrm{i}) = \frac{\pi}{4}$, so on average each step stretches the iterates of $\mathbf{B}$ by $\sqrt{2}$ and turns them through $\frac{\pi}{4}$: they spiral outwards. After four steps they are back on their own line, reversed: $(1+\mathrm{i})^4 = -4$, and indeed $\mathbf{B}^4 = -4\mathbf{I}$.

Check. $\mathbf{A}(0,2,1)^{\mathsf{T}} = (0,6,3)^{\mathsf{T}} = 3(0,2,1)^{\mathsf{T}}$; $\operatorname{tr}\mathbf{A} = 6 = 1+2+3$ and $\det\mathbf{A} = 6 = 1\cdot 2\cdot 3$. In (b), $\mathbf{B}(5, 3-\mathrm{i})^{\mathsf{T}} = (5+5\mathrm{i},\ 4+2\mathrm{i})^{\mathsf{T}} = (1+\mathrm{i})(5, 3-\mathrm{i})^{\mathsf{T}}$; $\operatorname{tr}\mathbf{B} = 2$ and $\det\mathbf{B} = 2 = (1+\mathrm{i})(1-\mathrm{i})$.

Pitfalls

- Subtracting λ from one diagonal entry only, or from every entry. It comes off each diagonal entry and nowhere else.
- Row-reducing $\mathbf{A} - \lambda\mathbf{I}$ to a matrix with no zero row. Then the only solution is $\mathbf{0}$, and λ is not an eigenvalue: recheck the characteristic polynomial.
- Keeping the same eigenvector for both halves of a complex pair. The eigenvector of $1 - \mathrm{i}$ is $(5, 3+\mathrm{i})^{\mathsf{T}}$, not $(5, 3-\mathrm{i})^{\mathsf{T}}$.

In the notes. Unit 6: Finding eigenvalues and eigenvectors, with its trace and determinant checks, and Unit 6: Complex eigenvalues.

Practise it. the Review set on this method

6.2 Is it diagonalisable? Multiplicities

Problem. Decide whether a matrix is diagonalisable, giving the algebraic and geometric multiplicity of each eigenvalue.

Method

1. Find the eigenvalues (Method 6.1), and the *algebraic multiplicity* of each: how many times it repeats as a root. A triangular matrix has its diagonal entries as eigenvalues.
2. If the n eigenvalues are distinct, the matrix is diagonalisable. Stop.
3. For each repeated eigenvalue, solve $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ and count the free unknowns: that is its *geometric multiplicity*, the number of independent eigenvectors.
4. Compare. If geometric = algebraic for every eigenvalue, the matrix is diagonalisable. If any eigenvalue falls short, it is defective and cannot be diagonalised.

Why it works. An $n \times n$ matrix is diagonalisable exactly when it has n independent eigenvectors. Eigenvectors of distinct eigenvalues are independent, and the geometric multiplicity never exceeds the algebraic one, so a simple root (one that occurs once) always gives exactly one eigenvector. Only a repeated root can fall short, and a real symmetric matrix never does.

Example. Are these diagonalisable?

$$\mathbf{A} = \begin{pmatrix} 4 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 2 & 3 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 4 & 0 & 0 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{pmatrix}.$$

Expand along row 1. The two matrices differ only in the $(3,1)$ entry, which the expansion never uses, so both have

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (4-\lambda)\big[(2-\lambda)(3-\lambda) - 2\big] = (4-\lambda)(\lambda-1)(\lambda-4).$$

So $\lambda = 4$ has algebraic multiplicity 2, and the simple root $\lambda = 1$ gives one eigenvector. For $\lambda = 4$,

$$\mathbf{A} - 4\mathbf{I} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & -2 & 1 \\ -1 & 2 & -1 \end{pmatrix}, \qquad \mathbf{B} - 4\mathbf{I} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & -2 & 1 \\ 1 & 2 & -1 \end{pmatrix}.$$

For $\mathbf{A}$, row 3 is -1 times row 2, so the one equation $v_1 - 2v_2 + v_3 = 0$ leaves two unknowns free, with eigenvectors $(1,0,-1)^{\mathsf{T}}$ and $(0,1,2)^{\mathsf{T}}$:

$$\mathbf{A}: \quad \boxed{\lambda = 4: \text{ alg. } 2, \text{ geom. } 2} \quad \text{diagonalisable.}$$

For $\mathbf{B}$, $R_3 - R_2 = (0,4,-2)$ gives $v_3 = 2v_2$, and then row 2 gives $v_1 = 0$. One unknown is free, and the only direction is $(0,1,2)^{\mathsf{T}}$:

$$\mathbf{B}: \quad \boxed{\lambda = 4: \text{ alg. } 2, \text{ geom. } 1} \quad \text{defective.}$$

Check. $\mathbf{A}(1,0,-1)^{\mathsf{T}} = (4,0,-4)^{\mathsf{T}}$ and $\mathbf{A}(0,1,2)^{\mathsf{T}} = (0,4,8)^{\mathsf{T}}$, both 4 times the vector. And $\operatorname{tr}\mathbf{A} = 9 = 4+4+1$, $\det\mathbf{A} = 16 = 4\cdot 4\cdot 1$. For $\mathbf{B}$, the first fails: $\mathbf{B}(1,0,-1)^{\mathsf{T}} = (4,0,-2)^{\mathsf{T}}$.

Pitfalls

- Calling every repeated eigenvalue a failure. $\mathbf{A}$ has $\lambda = 4$ twice and is diagonalisable; only the eigenspace decides.
- Counting rows instead of free unknowns. Row-reduce first: $\mathbf{B} - 4\mathbf{I}$ keeps two non-zero rows in three unknowns, so one unknown is free (geometric multiplicity 1); $\mathbf{A} - 4\mathbf{I}$ keeps one, so two are free.
- Working hard on the simple eigenvalue. A root of multiplicity 1 always has geometric multiplicity 1; check the repeated roots.

In the notes. Unit 6: Diagonalisation, for the criterion, and Unit 6: When diagonalisation fails, for the multiplicities.

Practise it. the Review set on this method

6.3 Diagonalising, and powers of a matrix

Problem. For a diagonalisable 2×2 matrix (Method 6.2), write $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$, and use it to find a closed form for $\mathbf{A}^k$.

Method

1. Find the eigenvalues and an eigenvector for each (Method 6.1).
2. Put the eigenvectors in the columns of $\mathbf{P}$, and the eigenvalues on the diagonal of $\mathbf{D}$, in the same order.
3. Compute $\det \mathbf{P}$, and $\mathbf{P}^{-1}$ by the 2×2 formula: swap the diagonal entries, negate the other two, divide by $\det \mathbf{P}$.
4. Then $\mathbf{A}^k = \mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}$, with $\mathbf{D}^k = \operatorname{diag}(\lambda_1^k, \lambda_2^k)$.
5. Multiply $\mathbf{P}\mathbf{D}^k$ first, then multiply the result by $\mathbf{P}^{-1}$.
6. Check that $k = 1$ gives $\mathbf{A}$ back.

Example. Diagonalise $\mathbf{A} = \begin{pmatrix} -1 & 3 \\ -6 & 8 \end{pmatrix}$ and find a closed form for $\mathbf{A}^k$.

$\det(\mathbf{A}-\lambda\mathbf{I}) = (-1-\lambda)(8-\lambda)+18 = \lambda^2-7\lambda+10 = (\lambda-2)(\lambda-5)$, so $\lambda_1 = 2$ and $\lambda_2 = 5$. For $\lambda_1 = 2$, $\mathbf{A}-2\mathbf{I} = \begin{pmatrix} -3 & 3 \\ -6 & 6 \end{pmatrix}$ gives $v_1 = v_2$: $(1,1)^{\mathsf{T}}$. For $\lambda_2 = 5$, $\mathbf{A}-5\mathbf{I} = \begin{pmatrix} -6 & 3 \\ -6 & 3 \end{pmatrix}$ gives $v_2 = 2v_1$: $(1,2)^{\mathsf{T}}$. With $\det \mathbf{P} = 2-1 = 1$,

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}, \quad \mathbf{P}^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}.$$

Then $\mathbf{P}\mathbf{D}^k = \begin{pmatrix} 2^k & 5^k \\ 2^k & 2\cdot 5^k \end{pmatrix}$, and multiplying by $\mathbf{P}^{-1}$,

$$\boxed{\mathbf{A}^k = \begin{pmatrix} 2^{k+1}-5^k & 5^k-2^k \\ 2^{k+1}-2\cdot 5^k & 2\cdot 5^k-2^k \end{pmatrix}}$$

Check. At $k = 1$: $\begin{pmatrix} 4-5 & 5-2 \\ 4-10 & 10-2 \end{pmatrix} = \mathbf{A}$. At $k = 2$ the formula gives $\begin{pmatrix} -17 & 21 \\ -42 & 46 \end{pmatrix}$, which is $\mathbf{A}\mathbf{A}$ multiplied out.

Variants. For one eigenvector no $\mathbf{P}$ is needed: $\mathbf{A}^k\mathbf{v} = \lambda^k\mathbf{v}$. The same product with $\mathbf{D}^{-1} = \operatorname{diag}(1/\lambda_1, 1/\lambda_2)$ gives $\mathbf{A}^{-1}$, when no eigenvalue is 0.

Pitfalls

- Putting the eigenvalues in $\mathbf{D}$ in a different order from the columns of $\mathbf{P}$. Swapping one without the other gives a wrong $\mathbf{A}$.
- Multiplying in the wrong order: $\mathbf{P}^{-1}\mathbf{D}^k\mathbf{P}$ is not $\mathbf{A}^k$. Check $k = 1$ to catch it.
- Raising the entries of $\mathbf{A}$ to the power. Only a diagonal matrix powers entry by entry: $\mathbf{A}^2$ has $(1,1)$ entry -17, not $(-1)^2 = 1$.

In the notes. Unit 6: Diagonalisation, and Unit 6: Computing powers, with its 2×2 example.
Practise it. the Review set on this method

6.4 A recurrence, by decomposing the start vector

Problem. Find a closed form for a sequence given by a two-term linear recurrence, or for $\mathbf{x}_n$ when $\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n$, when the 2×2 step matrix has two distinct eigenvalues.

Method

1. Write the step as a matrix, $\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n$, so that $\mathbf{x}_n = \mathbf{A}^n\mathbf{x}_0$. For a two-term recurrence, stack two consecutive terms, the later one on top.
2. Find the eigenpairs $\lambda_1, \mathbf{v}_1$ and $\lambda_2, \mathbf{v}_2$ (Method 6.1).
3. Decompose the start: solve $\mathbf{x}_0 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ for c_1 and c_2.
4. Each eigenvector is scaled by its eigenvalue at every step: $\mathbf{x}_n = c_1\lambda_1^n\mathbf{v}_1 + c_2\lambda_2^n\mathbf{v}_2$.
5. Read off the component asked for.
6. Check a few small n against the recurrence. Only if the long run is asked for: when $|\lambda_1| > |\lambda_2|$ strictly and $c_1 \neq 0$, the λ_1 term dominates.

Example. A sequence has $x_{n+1} = x_n + 6x_{n-1}$, with $x_0 = 3$ and $x_1 = 4$. (a) Find a closed form for x_n. (b) What does x_{n+1}/x_n tend to?

(a) Stack two terms: the rows $x_{n+1} = x_n + 6x_{n-1}$ and $x_n = x_n$ give

$$\begin{pmatrix} x_{n+1} \\ x_n \end{pmatrix} = \begin{pmatrix} 1 & 6 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix}, \qquad \begin{pmatrix} x_{n+1} \\ x_n \end{pmatrix} = \mathbf{A}^n \begin{pmatrix} 4 \\ 3 \end{pmatrix}.$$

$\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 - \lambda - 6 = (\lambda - 3)(\lambda + 2)$. For $\lambda_1 = 3$, $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -2 & 6 \\ 1 & -3 \end{pmatrix}$ gives $v_1 = 3v_2$: $(3, 1)^{\mathsf{T}}$. For $\lambda_2 = -2$, $\mathbf{A} + 2\mathbf{I} = \begin{pmatrix} 3 & 6 \\ 1 & 2 \end{pmatrix}$ gives $v_1 = -2v_2$: $(-2, 1)^{\mathsf{T}}$. Decompose the start:

$$\begin{pmatrix} 4 \\ 3 \end{pmatrix} = c_1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} : \quad 3c_1 - 2c_2 = 4, \quad c_1 + c_2 = 3,$$

so $c_1 = 2$ and $c_2 = 1$. Scaling each eigenvector by its eigenvalue,

$$\begin{pmatrix} x_{n+1} \\ x_n \end{pmatrix} = 2 \cdot 3^n \begin{pmatrix} 3 \\ 1 \end{pmatrix} + (-2)^n \begin{pmatrix} -2 \\ 1 \end{pmatrix},$$

and the second component is

$$\boxed{x_n = 2 \cdot 3^n + (-2)^n}$$

(b) $|3| > |-2|$ and $c_1 = 2 \neq 0$, so the first term wins: $x_n = 3^n\left(2 + (-\tfrac{2}{3})^n\right)$, and $\boxed{x_{n+1}/x_n \to 3}$.

Check. The recurrence gives $x_2 = 4 + 18 = 22$ and $x_3 = 22 + 24 = 46$; the formula gives $18 + 4 = 22$ and $54 - 8 = 46$.

Pitfalls

- Putting x_0 on top of the start vector. The top entry is the later term, so the start is $(x_1, x_0)^{\mathsf{T}} = (4, 3)^{\mathsf{T}}$.
- Reading off the wrong component: the top one is x_{n+1}, the bottom one x_n.
- Claiming a limit when the eigenvalues tie in modulus. With $\lambda = \pm 2$, $x_n = 2^n\left(c_1 + (-1)^n c_2\right)$, and unless c_1 or c_2 is 0 the ratio x_{n+1}/x_n never settles: it alternates between two values, or every other term is 0.

In the notes. Unit 6: Computing powers, with Fibonacci and Binet's formula, and Unit 6: Why this chain converges, for dominance.

Practise it. the Review set on this method

6.5 Symmetric and Hermitian matrices: orthogonal diagonalisation, PCA

Problem. For a real symmetric or a Hermitian matrix with distinct eigenvalues, find its real eigenvalues and orthogonal eigenvectors; for a real symmetric one, also an orthogonal $\mathbf{Q}$ with $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}}$, and its principal axes.

Method

1. Check the structure: $\mathbf{A}^{\mathsf{T}} = \mathbf{A}$ for a real matrix; $\mathbf{H}^{\dagger} = \mathbf{H}$ for a complex one (a real diagonal, and conjugates across it).
2. Solve the characteristic equation. For a Hermitian matrix the off-diagonal product is $z\overline{z} = |z|^2$. The eigenvalues come out real.
3. Find an eigenvector for each as in Method 6.1: for a 2×2 from one non-zero row, choosing v_2 to clear any fraction; for a 3×3 by row-reducing $\mathbf{A} - \lambda\mathbf{I}$.
4. Check that the eigenvectors are orthogonal: each pair has dot product 0. For complex vectors use $\mathbf{u}^{\dagger}\mathbf{v} = \overline{u_1}v_1 + \overline{u_2}v_2$, not the plain product.
5. For a real symmetric matrix, divide each eigenvector by its length. These are the columns of $\mathbf{Q}$, with the eigenvalues in $\mathbf{D}$ in the same order. Then $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}$ and $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}}$.
6. PCA: for a covariance matrix, the eigenvector of the largest eigenvalue is the first principal axis. Each eigenvalue measures the variance along its axis, and they sum to $\operatorname{tr}\mathbf{A}$, the total variance, so that axis carries the share $\lambda/\operatorname{tr}\mathbf{A}$.

Example. (a) Orthogonally diagonalise the covariance matrix $\mathbf{A} = \begin{pmatrix} 3 & 2 \\ 2 & 6 \end{pmatrix}$, and give its first principal axis and the share of the variance it carries. (b) Find the eigenvalues and eigenvectors of $\mathbf{H} = \begin{pmatrix} 2 & 1+2\mathrm{i} \\ 1-2\mathrm{i} & -2 \end{pmatrix}$, and check that the eigenvectors are orthogonal.

(a) $\mathbf{A}^{\mathsf{T}} = \mathbf{A}$. $\det(\mathbf{A} - \lambda\mathbf{I}) = (3-\lambda)(6-\lambda) - 4 = \lambda^2 - 9\lambda + 14 = (\lambda-2)(\lambda-7)$, so $\lambda_1 = 2$, $\lambda_2 = 7$. Row 1 reads $v_1 + 2v_2 = 0$ for λ_1, giving $(2,-1)^{\mathsf{T}}$, and $-4v_1 + 2v_2 = 0$ for λ_2, giving $(1,2)^{\mathsf{T}}$. Their dot product is $2 - 2 = 0$, and both have length $\sqrt{5}$:

$$\boxed{\mathbf{Q} = \frac{1}{\sqrt{5}}\begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}}, \qquad \boxed{\mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 7 \end{pmatrix}}$$

The first principal axis is $\frac{1}{\sqrt{5}}(1,2)^{\mathsf{T}}$, the eigenvector of the larger eigenvalue 7. The total variance is $\operatorname{tr}\mathbf{A} = 9$, so it carries $\boxed{\frac{7}{9} \approx 77.8\%}$.

(b) Conjugating $\mathbf{H}$ and transposing returns $\mathbf{H}$, so $\mathbf{H}^{\dagger} = \mathbf{H}$. With $(1+2\mathrm{i})(1-2\mathrm{i}) = 5$,

$$\det(\mathbf{H} - \lambda\mathbf{I}) = (2-\lambda)(-2-\lambda) - 5 = \lambda^2 - 9,$$

so $\lambda = \pm 3$, both real. For $\lambda = 3$ row 1 reads $-v_1 + (1+2\mathrm{i})v_2 = 0$; take $v_2 = 1$. For $\lambda = -3$ it reads $5v_1 + (1+2\mathrm{i})v_2 = 0$; take $v_2 = 5$ to clear the fraction:

$$\boxed{\lambda = 3 : \begin{pmatrix} 1+2\mathrm{i} \\ 1 \end{pmatrix}}, \qquad \boxed{\lambda = -3 : \begin{pmatrix} -1-2\mathrm{i} \\ 5 \end{pmatrix}}$$

The conjugate product is $(1-2\mathrm{i})(-1-2\mathrm{i}) + 1\cdot 5 = -5 + 5 = 0$: orthogonal.

Check. $\mathbf{A}(1,2)^{\mathsf{T}} = (7,14)^{\mathsf{T}} = 7(1,2)^{\mathsf{T}}$, and $\mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}} = \frac{1}{5}\begin{pmatrix} 15 & 10 \\ 10 & 30 \end{pmatrix} = \mathbf{A}$. For $\mathbf{H}$: $\operatorname{tr}\mathbf{H} = 0 = 3 + (-3)$

and $\det \mathbf{H} = -4 - 5 = -9 = 3 \cdot (-3)$.

Pitfalls

- Using the plain product for complex vectors. In (b) it gives $(1+2\mathrm{i})(-1-2\mathrm{i}) + 5 = 8 - 4\mathrm{i}$, which says nothing about orthogonality.
- Forgetting to normalise. With the columns $(2,-1)^{\mathsf{T}}$ and $(1,2)^{\mathsf{T}}$ as they stand, $\mathbf{P}^{\mathsf{T}}\mathbf{P} = 5\mathbf{I}$, so $\mathbf{P}^{-1} \neq \mathbf{P}^{\mathsf{T}}$.
- Taking the principal axis from the largest diagonal entry. The 6 in $\mathbf{A}$ suggests $(0,1)^{\mathsf{T}}$, but the axis is $\frac{1}{\sqrt{5}}(1,2)^{\mathsf{T}}$.

In the notes. Unit 6: Symmetric and Hermitian matrices, with the note on the conjugate inner product, and Unit 6: Symmetric matrices and orthogonal diagonalisation.

Practise it. the Review set on this method

6.6 A Markov chain's long run

Problem. Set up the transition matrix of a Markov chain; find its steady state, a closed form for $\mathbf{x}_n$, and whether the chain settles.

Method

1. Build $\mathbf{P}$: column j lists the probabilities of moving *from* state j. Check that each column sums to one.
2. $\lambda_1 = 1$ always. Find λ_2 from the characteristic equation; for a 2×2, $\lambda_1 + \lambda_2 = \operatorname{tr}\mathbf{P}$ checks it, and $\lambda_2 = 1$ only for $\mathbf{P} = \mathbf{I}$, where $\mathbf{x}_n = \mathbf{x}_0$: stop.
3. Steady state: solve $(\mathbf{P} - \mathbf{I})\mathbf{x} = \mathbf{0}$, scale $\mathbf{x}$ so its entries sum to one, and check $\mathbf{P}\mathbf{x}_\infty = \mathbf{x}_\infty$.
4. Closed form, for two states: find the eigenvector $\mathbf{v}_2$ of λ_2, and write $\mathbf{x}_0 = c_1\mathbf{x}_\infty + c_2\mathbf{v}_2$. Since $\mathbf{v}_2$ sums to zero, $c_1 = 1$ for every probability start, and $\mathbf{x}_n = \mathbf{x}_\infty + c_2\lambda_2^n\mathbf{v}_2$.
5. Long run: the chain settles to $\mathbf{x}_\infty$ from every start when $\lambda = 1$ occurs just once and every other eigenvalue has $|\lambda| < 1$. A second eigenvalue of modulus 1 (such as -1) can keep it moving for ever.

Example. Each day a machine is either working or under repair. A working machine is still working the next day with probability 0.9; one under repair is working again the next day with probability 0.3. It starts under repair. Find the steady state, and the probability that it is working on day n.

With $\mathbf{x}_n = (\text{working},\ \text{under repair})$, the columns are the two from-states:

$$\mathbf{P} = \begin{pmatrix} 0.9 & 0.3 \\ 0.1 & 0.7 \end{pmatrix}, \qquad \mathbf{x}_0 = (0,\ 1).$$

Both columns sum to one. $\det(\mathbf{P} - \lambda\mathbf{I}) = \lambda^2 - 1.6\lambda + 0.6 = (\lambda - 1)(\lambda - 0.6)$, so $\lambda_1 = 1$ and $\lambda_2 = 0.6$, and $1 + 0.6 = 1.6 = \operatorname{tr}\mathbf{P}$. For $\lambda_1 = 1$, $(\mathbf{P} - \mathbf{I})\mathbf{x} = \mathbf{0}$ gives $-0.1x_1 + 0.3x_2 = 0$, so $x_1 = 3x_2$: the direction $(3,1)$. Scaled to sum one,

$$\boxed{\mathbf{x}_\infty = (0.75,\ 0.25)}$$

For $\lambda_2 = 0.6$, $\mathbf{P} - 0.6\mathbf{I} = \begin{pmatrix} 0.3 & 0.3 \\ 0.1 & 0.1 \end{pmatrix}$ gives $(1, -1)$. Then $\mathbf{x}_0 = (0, 1) = (0.75, 0.25) + c_2(1, -1)$ gives $c_2 = -0.75$, so

$$\mathbf{x}_n = (0.75,\ 0.25) - 0.75\,(0.6)^n\,(1, -1),$$

and the probability of working on day n is

$$\boxed{0.75 - 0.75\,(0.6)^n}$$

$\lambda = 1$ occurs once and $|0.6| < 1$, so this tends to 0.75 from any start.

Check. $\mathbf{P}(0.75, 0.25) = (0.675 + 0.075,\ 0.075 + 0.175) = (0.75, 0.25)$. One step from the start, $\mathbf{P}\mathbf{x}_0 = (0.3, 0.7)$, and the formula gives $0.75 - 0.45 = 0.3$.

Pitfalls

- Putting the from-states in the rows. That builds $\mathbf{P}^{\mathsf{T}}$, whose columns here sum to 1.2 and 0.8: the column check catches it.
- Giving the direction $(3, 1)$ as the steady state. A distribution sums to one: $(0.75, 0.25)$.
- Promising convergence without checking the other eigenvalues. The swap $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has eigenvalues ± 1 and oscillates for ever, and $\mathbf{I}$ has $\lambda = 1$ twice and never forgets its start.

In the notes. Unit 6: Markov chains, Unit 6: The steady state is an eigenvector, and Unit 6: Why this chain converges.

Practise it. the Review set on this method